David Danielian

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SUMMARY

Full-stack engineer building 3D reconstruction and spatial data systems, with a robotics background.

WORK EXPERIENCE

Smartapp.com Worcester, MA Sept 2023 - Present

Full Stack Engineer

- Developed and deployed a scalable, containerized spatial reconstruction pipeline leveraging Gaussian Splatting to create digital twins of indoor spaces from video input.
- Architected pipeline end to end, including Python code, designing SQL and No-SQL databases, Docker containerization, and job orchestration via Kubernetes and Argo Workflows.
- Developed ROS 2 Nodes for SLAM, A* path planning, obstacle detection/avoidance, and odometry for prototype robotics platform on Jetson Hardware with Realsense stereo pair
- Set up CI/CD workflows including automated test suites, and automated builds and cloud function deployment via Jenkins, allowing for easy weekly releases.
- Built dashboards with React/Typescript to monitor per-job costs and pipeline metrics in real-time.
- Configured GKE clusters, cross-project IAM, Workload Identity, API Secret Management, and Firestore-based distributed locking to support reliable concurrent pipeline execution.

Catchpoint Remote May 2022 - Aug 2022

Software Engineering Intern

- Updated legacy web pages written in .NET Web Forms to React GraphQL stack

Human-Computer Interaction Lab WPI Worcester, MA May 2021 - May 2023

Research Software Developer

- Built web-based fNIRS signal processing and analysis tools for neuroscience/ML research
- Implemented queueing and websockets to enable the preprocessing of 50+ GB datasets

PROJECTS

Brain-Computer Interface to Improve Educational Software

WPI Capstone project. Developed a Brain-Computer interface that uses ML models to monitor and visualize a user's cognitive state while doing learning tasks. Created Chrome extension with Flask API to receive brain data and display visuals in the browser.

VR Haptic Gloves

Developed VR haptics gloves with finger position tracking, and force feedback. Responsible for sensor analysis, writing firmware in C++, handling IPC, and creating test environments in Unity**.**

SKILLS

Cloud & Infrastructure: GCP, Docker, Kubernetes, Firestore, Cloud Functions, MySQL, AWS, Lambda
Web Development: HTML, CSS, Next.js, React, Flask, Three.js, Firebase
Programming Languages: Python, JavaScript/TypeScript, C#, SQL, Java, C++, C
Tools: Git, Linux, Jenkins, ROS, Unity, Godot, Figma

EDUCATION

Worcester Polytechnic Institute Worcester MA **Class of 2023**

Bachelors in Computer Science, Robotics Engineering Minor, 3.76 GPA

References are available upon request to protect my mentors and colleagues privacy